



Distortion Pedal With Stutter Effect

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Intro: Distortion Pedal With Stutter Effect

This was built for a friend who asked me for a distortion pedal for him to use with his guitar. The main distortion effect is achieved through an LM386 audio amplifier IC which clips the signal. There is also a stutter effect which turns the output signal on and off at a user-specified rate. This is controlled by one of the three potentiometers (the others control Gain and Output Volume). An NE555 timer IC generates a square wave which turns the amplifier section on and off. The 555 oscillator circuitry and the original idea are based on [a design by spel3o](#).

The stutter effect isn't to everybody's taste, so I added a toggle switch to switch it out of the circuit. A bit of advice - the NE555 timer generates a fair bit of noise and pops on the power supply, so I wired the switch to also remove power from the NE555.



Step 1: Parts list

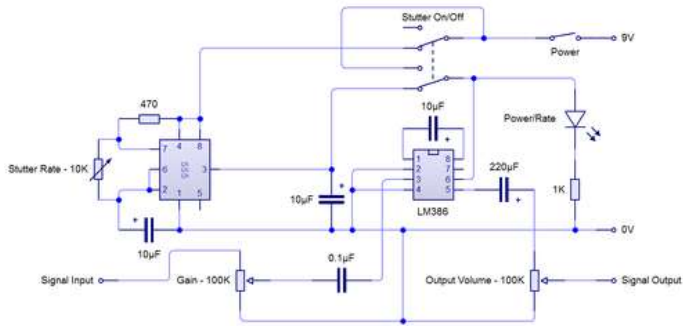
The following components and parts were used in this project :

- (1x) Die-cast aluminium enclosure - I used the Hammond 29830PSLA
- (3x) Potentiometer control knobs
- (1x) Heavy-duty latching DPDT push switch
- (2x) DPDT miniature toggle switches
- (1x) 2.1mm DC power socket (Use a plastic body one if you are using a metal enclosure!)
- (2x) 1/4" jack sockets
- (1x) 5mm LED bezel
- (1x) Battery snap
- (1x) PP3 9V battery
- (1x) Piece of stripboard
- (2x) 8 pin DIL sockets
- (1x) 10K potentiometer
- (2x) 100K potentiometers
- (1x) 470R resistor
- (1x) 1K resistor
- (3x) 10uF electrolytic capacitors
- (1x) 220uF electrolytic capacitor
- (1x) 0.1uF (100nF) polyester or ceramic capacitor
- (1x) 5mm LED of your favourite colour
- (1x) NE555 timer IC
- (1x) LM386 audio amplifier IC



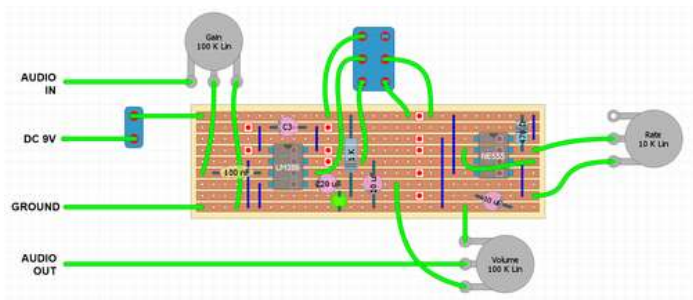
Step 2: Circuit diagram

Here is the schematic diagram for the circuit. If you choose not to build it on stripboard, you can build it according to this instead. To see an enlarged version of it, the link is here: <http://www.instructables.com/files/orig/F6Y/ODX1/GWF65G6U/F6YODX1GWF65G6U.png>



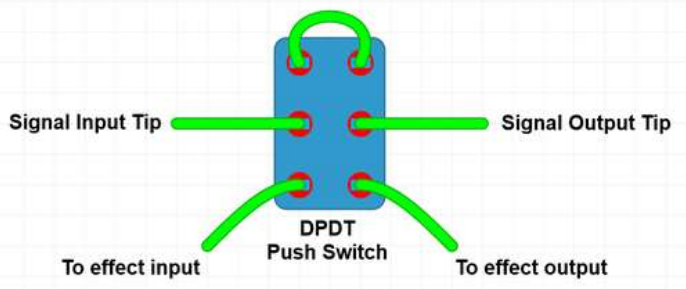
Step 3: Stripboard layout

In this diagram, the little red squares with dots in the middle represent track breaks. Also, the capacitor across pins 1 and 8 of the LM386 is a 10uF electrolytic - I forgot to label it. Watch out for the track breaks under components. To see an enlarged version of it, the link is here: <http://www.instructables.com/files/orig/FYR/9FLD/GWKKMP05/FYR9FLDGWKKMP05.png>



Step 4: Bypass switch

This is the wiring diagram for the DPDT heavy-duty push switch. It allows for true bypass - that is to say that when the effect is bypassed, the effect is completely removed from the effect chain so it won't steal any of your signal.



Step 5: Power jack

The power jack is wired so that when the AC adapter is connected, the internal 9V battery is automatically disconnected. The 9V supply going to the board is interrupted by a switch, which turns the pedal on or off.

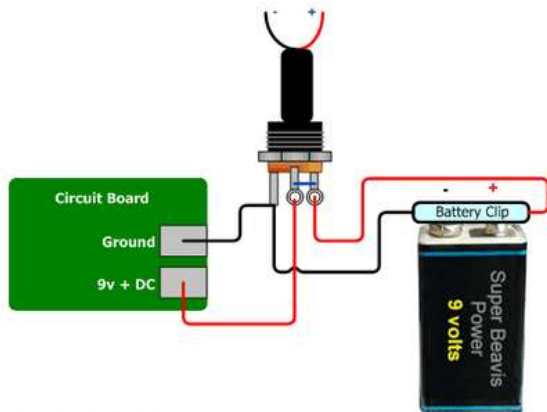
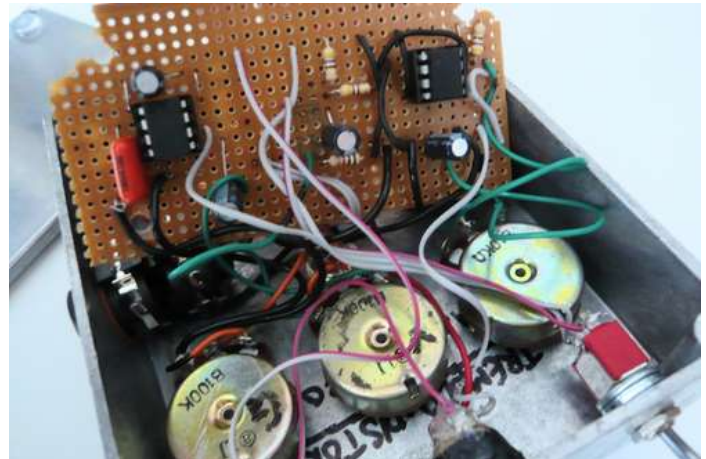
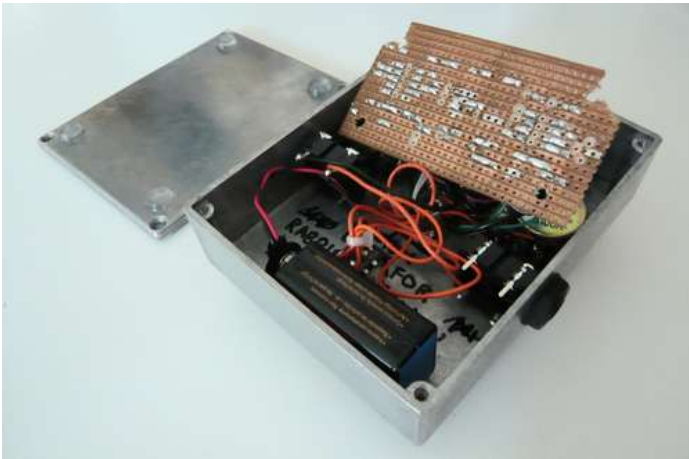


Diagram from Beavis Audio
<http://www.beavisaudio.com>

Step 6: Enclosure

As I mentioned before, I used a diecast aluminium Hammond 29830PSLA enclosure. This family of enclosures is the de-facto standard for DIY effect pedals, as these enclosures are very durable yet easy to machine. I decided to leave mine unfinished and unlabeled, as it makes it look a bit more interesting. I did, however, polish up the enclosure so it was nice and shiny. The layout is up to you, but the layout I decided on worked very well and everything fit neatly in the space.



Step 7: Final thoughts

The friend who I made this pedal for is very pleased with it. He promises he'll send a video of it in operation soon.

The pedal does have a couple of quirks however. There is a very slight pop when the signal is being turned on or off. This is not a problem most of the time as you cannot hear it when sound is being played, but when the rate is set to highest, the pops from turning on the signal are so high in frequency that they become an oscillator and produce an audible tone. There isn't really any way to eliminate this without completely redesigning the circuit.

Also, it would have been nice to sand the enclosure with wet & dry paper as a chrome finish would have looked fantastic.

All in all, this was a fun project to make and the new owner is very happy with the effect. Thank you to spel3o for the original circuit on which this project is based.



Related Instructables



Make Your Own Tremolo Effects Pedal by spel3o



Helicopter - Auto Stutter Stompbox (Photos) by jakebaldwin



Simple overdrive effect pedal by magnus0



Boss TR2 C4 Mod by raxel



Create Your Own Guitar Effects Pedal by themakeclass



The ValveLiTzer: Low-voltage Tube Booster by gmoon